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Gate bias modulation towards organic electrochemical transistors with ultra-high cycling stability†

Miao Xie,[a] Yueping Lai,[b] Meisi Li,[c] Dan Zhao,[a] Chenggeng Huang,[a] Libing Bai,[a] Yuhua Cheng,[a] Jianhua Chen,*[c] Liang-Wen Feng*[b] and Wei Huang *[a]

Organic electrochemical transistors (OECTs) show great potential in next-generation bioelectronics due to their high transconductance, low driving voltage, and biocompatibility. These advantages rely on efficient electrochemical doping/dedoping of the transistor channel consisting of organic mixed ionic–electronic conductors (OMIECs). Nevertheless, during device operation, hydration and electrolyte swelling lead to inferior cycling/operation stability of OECTs when compared with other conventional transistors. To enable further development and commercialization of OECTs, long-term stable operation is mandatory. Here, it is discovered that a slight variation in gate bias (V_G) during operation would result in contrasting degradation in OECT performance with a conventional planar structure based on p-type p(g2T-T), which is the result of the alternating consumption of active OMIEC content due to the swelling effect during cycling. The introduction of the vertical architecture along with a crosslinking agent in the channel would largely decelerate/avoid such a process, which could yield an OECT with more than 250 000 full cycles (cycling duration of 10 000 s). Moreover, this strategy can be extended to another n-type OMIEC (Homo-gDPP). This work reveals a unique class of bioelectronics where their lifetime can be managed and controlled by device structure engineering, showing extensive potential in biomedical and bio-implantable functional devices.

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Wei Huang

Wei Huang obtained his B. S. in physics from Nankai University and his PhD in optical engineering from University of Electronic Science and Technology of China (UESTC). He was a research assistant professor at Northwestern University, under the supervision of Professor Tobin J. Marks and Professor Antonio Facchetti. Currently, he is a Professor at the School of Automation Engineering, UESTC. His research interests include flexible/stretchable electronics and transistor-based chemical/biological sensors.

Introduction

Due to their low operational voltage, high transconductance (g_m), and good biocompatibility, organic electrochemical transistors (OECTs) have received substantial attention as efficient ionic-to-electronic transducers in the field of bioelectronics,[1–4] such as for electrophysiological signal recording,[5–7] cell monitoring,[8–10] and neuromorphic computing.[11–13] In an OECT, the core component is the organic mixed ionic–electronic (semi)conductor (OMIEC), which acts as the channel material in direct contact with an electrolyte.[14] Under the gate bias (V_G), the ions from the electrolyte can be injected into or ejected from the channel, effectively doping/dedoping the bulk of the OMIEC and modulating the drain current (I_D).[15] Owing to ultra-large volume capacitance, a small change in V_G can lead to substantial modulation in I_D, leading to high g_m (typically > mS) under low driving voltage (<1 V).[16–18]

The operation of OECTs relies on both ionic diffusion and charge carrier transport in OMIECs. Considering that current

[a]School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, P. R. China. E-mail: whuang@uestc.edu.cn

[b]Key Laboratory of Green Chemistry & Technology, Ministry of Education, College of Chemistry, Sichuan University, Chengdu 610065, P. R. China. E-mail: liangwenfeng@scu.edu.cn

[c]School of Chemical Science and Technology, Yunnan University, Kunming 650091, P. R. China. E-mail: chenjianhua@ynu.edu.cn

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modulation is a consequence of ionic doping/dedoping in the OMIEC channel, this process will inevitably result in irreversible variations in the microstructure of the channel. Specifically, the injection/ejection process of ions, particularly ions carrying large solvation shells, can lead to tremendous swelling of OMIEC, which may weaken the intermolecular interactions and entanglement of polymer chains.[19,20] Consequently, irreversible degradation of the OECT performance (typically I_D) would be observed after repeated cycling processes.[21] Note that one cycling test of an OECT is defined as the OMIEC is doped by ions (transistor turns on) and then dedoped (transistor turns off) by the modulation of a square wave V_G under a constant drain voltage (V_D). In addition, when an OECT operates in an aqueous electrolyte with the presence of oxygen, electrochemical side reactions such as the oxygen reduction reaction (ORR) are hard to avoid. ORR can yield either H_2O_2 through a two-electron process:[22,23]

$$O_2 + 2H^+ + 2e^- \rightarrow H_2O_2 \quad (1)$$

or yield H_2O in a four-electron process:

$$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O \quad (2)$$

The formation of H_2O_2 may cause corrosion of the OMIEC and decrease the cycling stability.[24]

Currently, OECTs with good cycling stabilities have been reported from the perspective of novel OMIECs. For instance, to suppress excessive swelling and water uptake, strategies such as side chain redistribution,[25] alkyl spacer introduction,[26] or side chain length shortening were employed.[27] As for ORR, degassing the electrolyte is the simplest method to inhibit ORR, but it is impractical to use oxygen-free electrolytes in actual applicable scenarios. Designing OMIECs with ionization potential (IP) $\geq$4.9 eV can partially avoid non-capacitive faradaic side reactions and prevent the formation of reactive side products such as H_2O_2.[24] Moreover, adding additives or dopants in OMIECs is also an effective method to improve the steady-state performance and stability of OECTs.[28–30] For example, (3-glycidyloxypropyl)trimethoxy silane (GOPS) additive could react with the sulfonic acid group of the excess poly(styrene sulfonate) in poly(3,4-ethylenedioxythiophene) : poly(styrene sulfonate) (PEDOT : PSS) channels, which led to a change in the channel morphology and effectively enhanced the cycling stability.[31] Additionally, the introduction of solid electrolytes such as ion gels and hydrogels has also demonstrated significant improvements in the cycling stability of OECTs.[32–36] Above all, advanced OMIEC synthesis, additive engineering, and solid electrolyte utilization have remarkably enhanced cycling stability, but OECTs with extremely stable cycling characteristics are still missing, which is indispensable for commercialization.[37–41]

In this work, by regulating the gate bias applied on OECTs with both planar and vertical structures, ultra-high cycling stability is achieved, where stable transfer characteristics are obtained after 250 000 cycles (a total of 10 000 s cycling time). Note, previous evaluations on the cycling stability of OECTs examine only the I_D level, while other key parameters, including g_m, and turn-on voltage (V_{ON}), were ignored. Here, significant variations of various key parameters of OECTs based on poly(2-(3,3′-bis(2-(2-(2-methoxyethoxy)ethoxy)ethoxy)-[2,2′-bithiophen]-5)yl thiophene) [p(g2T-T)] upon cycling tests under different V_G windows (+0.4 to −0.4 V, +0.4 to −0.5 V, and +0.4 to −0.6 V) are observed in OECTs with planar structure, where narrower V_G windows lead to better stability. By using atomic force microscopy (AFM) and cyclic voltammetry (CV), it is revealed that higher V_G accelerates the irreversible consumption of channel materials, showing rougher surface morphology and declining electrochemical active content. On the other hand, by adopting a vertical architecture, extremely stable transfer characteristics are obtained upon cycling, which is attributed to the sandwiched OMIEC between source/drain electrodes prohibiting the large contact area with the electrolyte, thus slowing down the irreversible consumption of the OMIEC channel. In addition, V_G modulation on the cycling stability can be further extended to another n-type OMIEC (Homo-gDPP) based OECT. This work reveals appropriate driving voltages combined with innovative structure design can contribute to a controllable and extended lifetime of functional bioelectronics.

Results and discussion

OECTs with planar architecture based on p(g2T-T) are first fabricated, and the cycling stabilities are evaluated under different V_G. The device schematic of planar OECT and chemical structures of p-type p(g2T-T) and crosslinker [double-end trifluoromethyl diazirines (DtFDA)],[42] are shown in Fig. 1a and S1.† Here, p(g2T-T) : DtFDA blend with a mass ratio of 4 : 1 is utilized as the channel material (see Experimental for the detailed fabrication process). The addition of DtFDA can introduce photo-patterning capability of the channel and maintain a relatively high content of p(g2T-T) to ensure high OECT performances. Cycling stability measurements are carried out for these OECTs by monitoring I_D over time upon applying a square wave of V_G (Fig. S2,† from +0.4 V to −0.4 V, from +0.4 V to −0.5 V or from +0.4 V to −0.6 V with the frequency of 25 Hz, respectively, V_D = −0.1 V). As shown in Fig. S3,† under different V_G conditions, the OECTs exhibit diverse degradation behaviours. Specifically, when V_G switches between +0.4 V and −0.6 V, significantly decreased I_D is observed, where maximum I_D in Fig. S3a† decreases from −0.18 mA (1st cycle) to −0.02 mA (100 000th cycle), remaining only 10.93% of the initial I_D. Such degradation is largely prohibited when the V_G pulse range is shrunk to a narrower range of +0.4 V to −0.5 V. As a result, 71.05% of the initial I_D is maintained after 100 000 cycles (−0.27 mA for the 1st cycle and −0.19 mA for the 100 000th cycle). More excitingly, when V_G switches between +0.4 V and −0.4 V, no apparent change in I_D over time is observed, where maximum I_D remains in a narrow range of −0.118 to −0.122 mA. However, in the characteristics of I_D over time, limited information can be accessed. Therefore, after every 5000 cycles, one transfer curve under consistent measurement procedure (V_G sweeping from +0.4 V to −0.6 V and then from −0.6 V to +0.4 V, V_D = −0.1 V) is acquired

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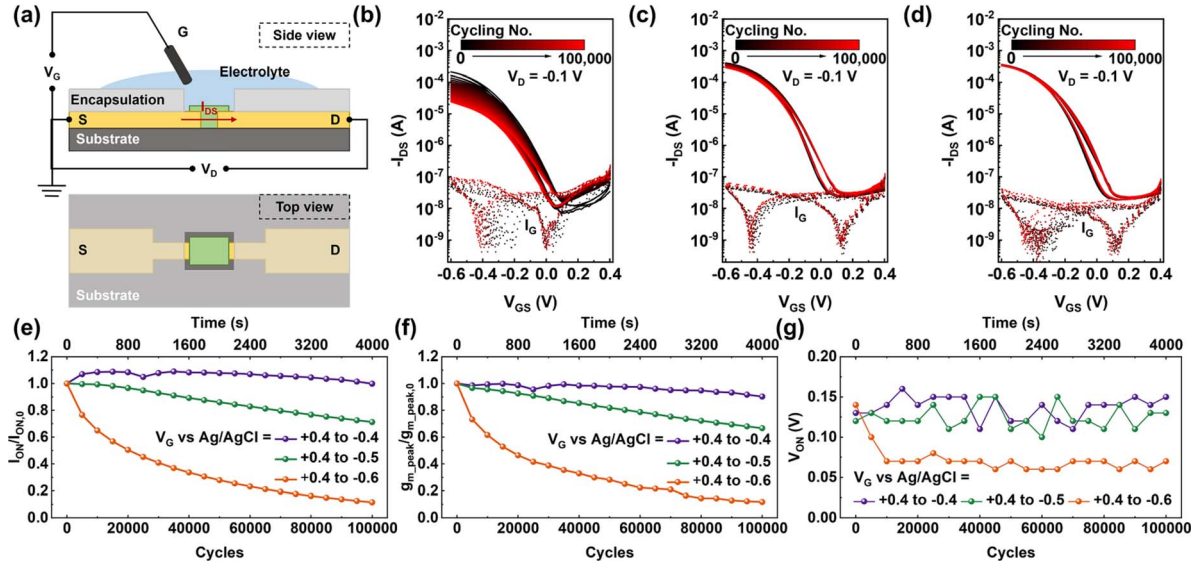


Fig. 1 (a) Device schematics of planar OECT. (b)–(d) Transfer characteristics of planar p(g2-T) : DtFDA OECTs during the cycling stability measurements with V_G switches from (b) +0.4 V to −0.6 V, (c) +0.4 V to −0.5 V and (d) +0.4 V to −0.4 V. Note, the V_G switching range indicated here refers to the voltage range of the square wave of V_G during cycling test, not the sweeping V_G range of the transfer curve. (e)–(h) Decay trends of (e) I_{ON}, (g) g_{m_peak} and (h) V_{ON}.

(Fig. 1b–d). The same V_G range in transfer curves is used to facilitate the comparison and quantification of the differences in cycling stability under varying gate biases during the cycling stability measurements. It is obvious that as the V_G window narrows down, the degradation trend of the transfer curves is inhibited. Especially for OECT cycled under V_G between +0.4 V and −0.4 V, almost identical transfer characteristics are obtained up to 100 000 cycles.

Variation trends of on-state current (I_{ON}, I_D under V_G = −0.6 V and V_D = −0.1 V), peak g_m (g_{m_peak}), and V_{ON} are extracted from the transfer curves and summarized in Fig. 1e–g and S4.† When V_G switches between +0.4 V and −0.6 V, I_{ON} and g_{m_peak} show a significant decline of 88.52% and 88.25%, with an apparent V_{ON} shift from +0.14 V to +0.07 V after 100 000 cycles. When the V_G window decreases by 0.1 V (between +0.4 V and −0.5 V), I_{ON} and g_{m_peak} decay 28.74% and 33.34%, respectively, with a relatively stable V_{ON} between +0.10 V and +0.15 V after 100 000 cycles. Further shrinking V_G window to +0.4 V to −0.4 V, stable I_{ON}, g_{m_peak}, and V_{ON} are observed, where only 0.20% and 9.69% of variations on I_{ON} and g_{m_peak} are observed after 100 000 cycles, along with stable V_{ON} between +0.11 V and +0.16 V. Moreover, with pure p(g2T-T) as the channel, similar degradation trends are still recorded (Fig. S5†). These results indicate that a narrower V_G window contributes to better cycling stability. Moreover, in all cases, the major variations in the device performance are I_{ON} and g_m, while no obvious changes in off-state current (I_{OFF}) and V_{ON} are observed. Therefore, it is suggested that during the cycling of planar OECTs, higher V_G would lead to a decrease in the active content of channel material, due to irreversible swelling and/or ORR side reactions.[21,24,25,43,44] The relatively constant I_{OFF} and V_{ON} in all cases, indicate that there is no obvious change in the property of the interface between source/drain electrodes and channel materials.[45] Note, for V_{ON}, planar OECT based on pure p(g2T-T) is an exception (Fig. S5†), where it monotonously shifts from 0.01 V (1st cycle) to −0.09 V (5000th cycle) under the V_G range of +0.4 V to −0.6 V, indicating that high V_G could affect the interface characteristics on the channel without crosslinker. It is worth noting that, the lack of errors in the above characteristic parameters is due to the requirement of a large amount of data for reasonable errors, which is time-consuming and unacceptable for laboratory-scale research. However, it is important to note that the above results are not accidental, as we have obtained consistent overall decay trend in multiple groups of data.

The above results suggest that high V_G could lead to irreversible decay due to fast consumption of active channel materials attributed to irreversible change on the microstructure after repeated swelling and/or electrochemical side reaction. To explore the relation between the gate bias and the cycling stability of OECT devices, the changes in the morphology of channel materials before and after the cycling stability measurements are compared by AFM.

As shown in Fig. 2a and b, the surface of the channel material becomes slightly rougher after 100 000 cycles of gate pulse [root-mean-square roughness (σ_{RMS}) = 1.27 nm and 1.89 nm for the pristine film and the cycled film, respectively]. Additionally, Fig. S6a and b† show a notable increase in σ_{RMS} of the semiconductor on the source electrode, with values of 1.60 nm and 2.78 nm. In contrast, σ_{RMS} of the semiconductor on the drain electrode does not change significantly before and after cycling, which are 1.30 nm and 1.50 nm, respectively (Fig. S6c and d†). It can be observed that under high V_G stress, ions repeatedly dope/dedope the materials on electrodes (especially on the source electrode) and in the channel, which irreversibly alters the microstructure of the material and is not

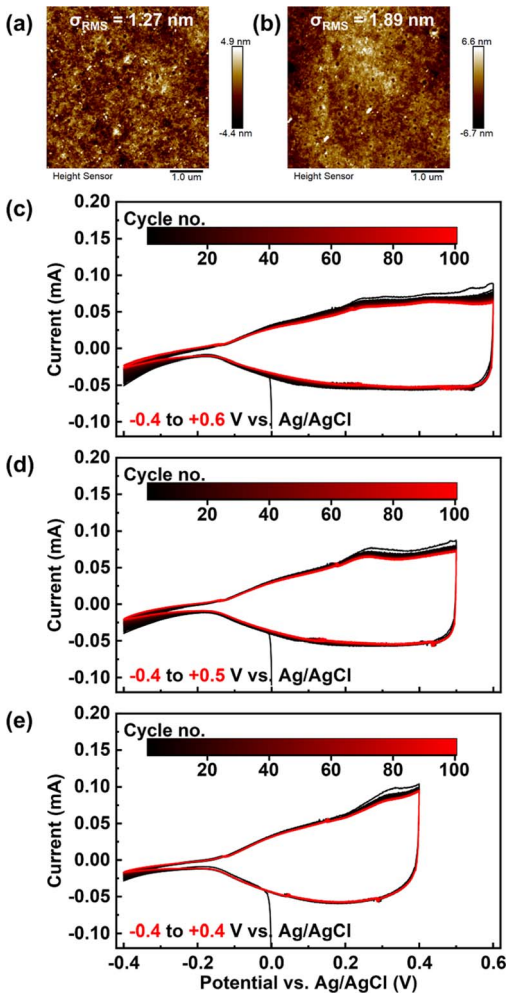


Fig. 2 (a) and (b) AFM images of the channel material (a) before and (b) after cycling stability measurement. (c)–(e) Cyclic voltammograms of the films cast from the p(g2T-T) : DtFDA mixture, with voltage steps of 1 mV (c) from −0.4 V to +0.6 V, (d) from −0.4 V to +0.5 V and (e) from −0.4 V to +0.4 V.

conducive to stable operation of OECTs. In addition, the effect of bias on the electrochemical behaviour of the film cast from the p(g2T-T) : DtFDA mixture is investigated. Solution of p(g2T-T) : DtFDA mixture is spin-coated on Cr (3 nm)/Au (50 nm) deposited glass substrates (5 mm × 5 mm) to act as the working electrode, while a saturated Ag/AgCl electrode acts as the reference electrode, and a 10 mm × 10 mm platinum plate is employed as the counter electrode. These samples are cycled in 0.01 M PBS solution using CV scans under different biases (from −0.4 V to +0.4 V, from −0.4 V to +0.5 V, or from −0.4 V to +0.6 V *versus* Ag/AgCl) with a scan rate of 50 mV s^{-1}. As shown in Fig. 2c–e, after 100 redox cycles, these films retain most of their CV signatures, but their stabilities differ. When the positive bias increases from +0.4 V to +0.5 V and +0.6 V, the current at the maximum potential decreases 8.64%, 16.36%, and 29.02%, respectively. For pure p(g2T-T), a similar trend is observed, where 9.69%, 14.53%, and 25.05% decreases in the maximum current are observed when cycling under −0.4 V to +0.4 V, −0.4 V to +0.5 V, and −0.4 V to +0.6 V, respectively (Fig. S7†). Therefore, it is speculated that higher gate bias accelerates the consumption of the channel material by hastening the mechanical disintegration of the OMIEC. When the gate bias increases, the higher degree of chemical oxidation (doping) of ions on the channel material, the ionic influx, and outflow cause more severe damage to the microstructure of the channel material, accelerating their irreversible consumption and reducing the cycling stability of OECTs.

To elaborate on this point, OECTs fabricated on a transparent glass substrate to improve contrast for easier observation of film morphology are further characterized. The morphological changes of the organic semiconductor thin film during long-term cycling stability testing are characterized using optical microscopy, as shown in Fig. S8† (V_G switches between +0.4 V and −0.6 V with a frequency of 25 Hz, $V_D = -0.1$ V). It can be seen that as the number of cycling cycles increases, the film morphology changes significantly. Initially, a smooth and uniform film covers the channel and the source/drain electrodes. While with continuing cycling, film on the source electrode side starts to reveal hole-like morphologies. After 100 000 cycles, a large area of the film covering the source electrode peels off. These observations suggest that during the long-term cycling stability testing, as the number of cycling cycles increases, the surface morphology of the OECT film gradually transitions from a uniform and smooth state to a roughened state, indicating irreversible damage to its microstructure, possibly due to the repeated doping/dedoping of ions during the swelling of the thin film. Furthermore, the bias stress on the source/drain electrodes and the presence of oxygen in the air contribute to the accelerated degradation of the film interface characteristic on the electrodes, particularly on the source electrode, leading to significant peeling of the thin film covering it.

On the other hand, compared to planar architecture, OECTs with vertical architecture are reported to hold higher g_m, short transient response time, and ultra-stable switching ability.[6,46–49] Device schematic of the vertical OECT (vOECT) is shown in Fig. 3a, where the same channel material [p(g2T-T) : DtFDA blend with a mass ratio of 4 : 1] is utilized. Cycling stability measurements under identical bias conditions are also conducted on vOECTs. Surprisingly, it is revealed that vOECTs exhibit notably superior cycling stability when compared to these of planar OECTs (Fig. S9†). After 100 000 cycles, the maximum I_D changes in Fig. S8a–c† are only 8.24% (from −11.62 mA to −12.58 mA), 2.61% (from −12.34 mA to −12.66 mA) and −0.75% (from −11.04 mA to −10.95 mA) under V_G of +0.4 V to-0.6 V, +0.4 V to −0.5 V, and +0.4 V to −0.4 V, respectively. Moreover, no significant differences are observed in the transfer characteristics of these devices (Fig. 3b–d). From the decay trends of I_{ON}, g_{m_peak}, and V_{ON} (Fig. 3e–g), it can be seen that there is no apparent change in the performance of vOECTs after 100 000 cycles. Specifically, in the case of V_G switches from +0.4 V to −0.4 V, the I_{ON} and g_{m_peak} show only 1.33% and 11.55% decrease, with a shift of V_{ON} from +0.33 V to +0.25 V after 100 000 cycles. When the gate cycling window increases by 0.1 V (from +0.4 V to −0.5 V), the I_{ON} and g_{m_peak} decay by 0.10% and 8.50%, respectively, with a shift of V_{ON} from +0.31 V to

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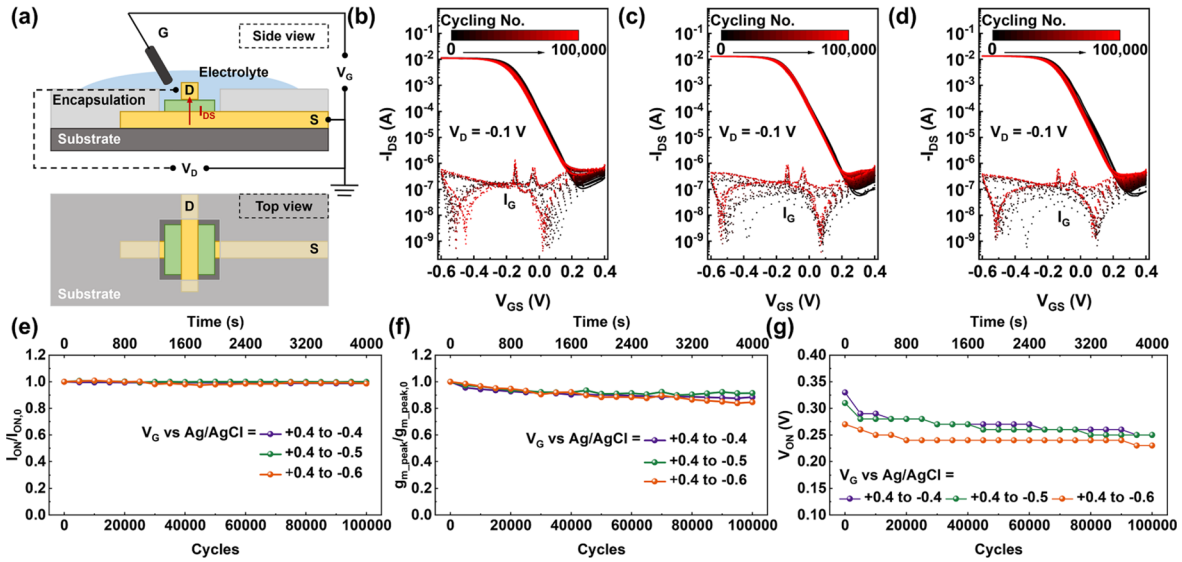


Fig. 3 (a) Device schematics of vertical OECT. (b)–(d) Transfer characteristics of vertical p(g2T-T) : DtFDA OECTs during cycling stability measurements with the V_G switches from (b) +0.4 V to −0.6 V, (c) +0.4 V to −0.5 V and (d) +0.4 V to −0.4 V. (e)–(g) Decay trends of (e) I_{ON}, (f) g_{m_peak} and (g) V_{ON}.

+0.25 V after 100 000 cycles. When the V_G switches from +0.4 V to −0.6 V, the I_{ON} and g_{m_peak} show an enhanced decline of 1.24% and 15.36%, respectively, with a shift of V_{ON} from +0.27 V to +0.23 V after 100 000 cycles. When compared with the degradation trends in planar OECTs (Fig. 1), it is obvious that vOECTs show superior cycling stability. Note, that I_{OFF} in vOECTs shows an obvious increase after cycling, but still remains at a relatively low level ($<10^{-6}$ A), which is slightly higher than that in planar OECTs. Such a phenomenon could be due to the short channel effect of vOECTs, where only ~100 nm channel length is utilized in these vOECTs. Moreover, vOECTs based on pure p(g2T-T) demonstrate poor cycling stability (Fig. S10†), since without a crosslinker in the channel, repeated channel swelling could lead to delamination between the top electrode (drain electrode) and the channel, and result in fast consuming of active p(g2T-T). These results indicate that the cycling stability of vOECTs is greatly enhanced by adding DtFDA as the crosslinker. Moreover, when compared to planar OECTs, vOECTs can effectively maintain their electrical performance when driven under different gate bias windows. Notably, as shown in Fig. S11,† vOECT can operate continuously for over 250 000 cycles (10 000 s) with a V_G switch from +0.4 V to −0.4 V, retaining a narrow I_D range of 101.86% to 110.98%.

The enhanced cycling stability of vOECTs under various gate biases can be attributed to the vertical stacking of the source and drain electrodes, which protects the channel material and reduces its contact area with the electrolyte, thereby slowing down material consumption and dissolution in the electrolyte. To further evaluate such speculation, redox stabilities of the p(g2T-T) : DtFDA films with different top covering Au layers are further analyzed by CV. Schematic diagrams of the samples (working electrode) are shown in Fig. 4a. These samples are cycled 100 cycles in 0.01 M PBS solution using CV scans with a fixed bias ranging from −0.4 V to +0.6 V *versus* Ag/AgCl (Fig. 4b–d). As expected, as the area of the top Au layer increases from 1.0 × 3.0 mm to 2.0 × 3.5 mm and 3.0 × 4.0 mm, the degradation trend of the maximum current is gradually inhibited, showing 24.00%, 21.03%, and 16.69% decreases, respectively. These results provide compelling evidence that the unique "sandwich" structure of vOECT plays a crucial role in ensuring long-term operational stability.

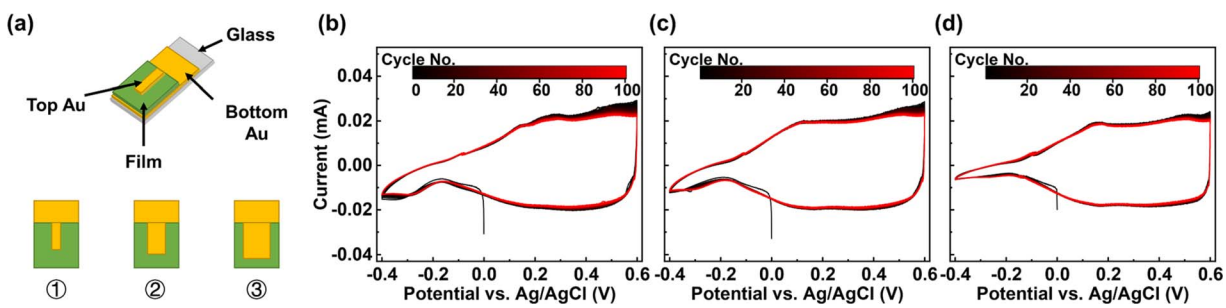


Fig. 4 (a) Schematic diagrams of CV measurement samples with varying sizes of the top layer of gold: ① 1.0 mm × 3.0 mm, ② 2.0 mm × 3.5 mm and ③ 3.0 mm × 4.0 mm. (b)–(d) Cyclic voltammograms of (b) sample ①, (c) sample ② and (d) sample ③.

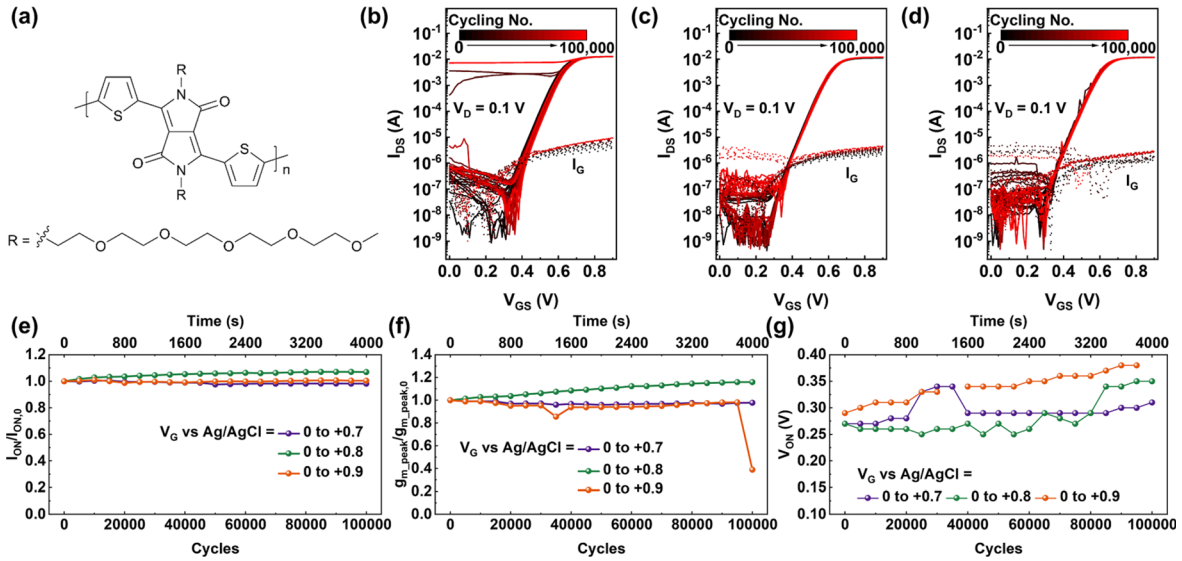


Fig. 5 (a) Chemical structure of Homo-gDPP. (b)–(d) Transfer characteristics of vertical Homo-gDPP : DtFDA OECTs during the cycling stability measurements with the V_G switches from (b) 0 V to +0.9 V, (c) 0 V to +0.8 V and (d) 0 V to +0.7 V. (e)–(g) Decay trends of (e) I_{ON}, (f) g_{m_peak} and (g) V_{ON}.

To further demonstrate the universal impact of gate bias on the cycling stability of OECTs, OECTs based on n-type Homo-gDPP (Fig. 5a) are fabricated and characterized along with DtFDA as crosslinker (Homo-gDPP : DtFDA is 3 : 1 in mass). The cycling stability measurements under different V_G bias conditions (V_G from 0 V to +0.7 V, from 0 V to +0.8 V, or from 0 V to +0.9 V with a frequency of 25 Hz) are also conducted in both planar and vertical architectures. With the support of vertical architecture, vOECTs based on Homo-gDPP : DtFDA mixture exhibit good cycling stabilities (up to 10 000 cycles) under different V_G biases (Fig. 5b–d and S12†), while the ones with planar structures show inferior stabilities (Fig. S13†). Specifically, when the V_G switches from 0 V to +0.7 V, the I_{ON} and g_{m_peak} show only a 1.84% and 2.24% decrease, with a shift of V_{ON} from +0.28 V to +0.31 V after 100 000 cycles. When the V_G switches from 0 V to +0.8 V, the I_{ON}, and g_{m_peak} increase 6.93%, and 15.93%, respectively after 100 000 cycles, with a shift of V_{ON} from +0.27 V to +0.35 V. While, when the V_G switches from 0 V to +0.9 V, the device can only operate stably for over 95 000 cycles. The above results indicate that using the vertical architecture can enhance the cycling stability of n-type OECTs under different gate bias conditions, but when the gate bias increases, it still leads to inferior cycling stability when compared to these under lower V_G bias.

Conclusions

In this work, the relation between V_G bias and the cycling stability of OECTs is systematically investigated along with planar and vertical architectures. By comparing the differences in cycling stability between planar and vertical OECTs under different V_G biases, it is found that the cycling stability of planar OECT is significantly affected by V_G bias, and the larger the gate bias range is, the poorer the cycling stability of the device is. In contrast, vOECTs exhibit good cycling stability in different V_G bias ranges. Through AFM, optical microscopy, and CV characterizations, it is demonstrated that higher V_G bias during the cycling of OECT would accelerate irreversible channel material consumption. While introduction of a vertical structure can effectively weaken the influence of V_G bias on the redox stability of channel film by protecting the active channel underneath the Au electrode, thereby improving the cycling stability of OECT. This study reveals the underlying reason for cycling stability degradation associated with V_G bias and provides a powerful architecture for the design of long-term stable operating bioelectronic devices.

Experimental

Materials

Semiconductive polymers p(g2T-T) and Homo-gDPP, along with crosslinkers DtFDA and Cinnamate-Cellulose (Cin-Cell), were synthesized based on previously reported procedures.[42,46,50,51] Isopropyl alcohol, acetone, and chloroform were purchased from Sigma-Aldrich and used as received without further purification. Micro-90 concentrated cleaning solution was obtained from International Products Corporation. Phosphate buffer saline (0.01 M PBS) supplied by Solarbio was used as the electrolyte. The Dow Corning Sylgard 184 silicone elastomer kit was utilized to prepare PDMS wells with a base/crosslinker mix ratio of 10 : 1.

Semiconductor solution preparation

First, p(g2T-T), Homo-gDPP, and Cin-Cell were dissolved in chloroform with concentrations of 10 mg mL^{-1}, 10 mg mL^{-1}, and 20 mg mL^{-1}, respectively. Additionally, DtFDA was dissolved in chloroform with two different concentrations of 20 mg

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mL^{-1} and 100 mg mL^{-1}. These solutions underwent filtration through a 0.45 μm polyvinylidene difluoride filter. Subsequently, the p(g2T-T) or Homo-gDPP solution was mixed with the DtFDA solution (20 mg mL^{-1}) in a mass ratio of 4 : 1 or 3 : 1, respectively. Cin-Cell solution and DtFDA solution (100 mg mL^{-1}) were mixed in a mass ratio of 4 : 1 to act as the precursor of the encapsulation layer.

Planar OECT fabrication

Si wafers with 100 nm of SiO_2 as substrates were first cleaned by ultrasonication in isopropyl alcohol for 15 minutes, followed by drying with N_2 and treating with UV-ozone (Shenzhen Hwo Technology Co., Ltd) for 15 minutes. When utilizing glass as substrates, the glass substrates underwent a similar cleaning process with ultrasonication in Micro-90 concentrated cleaning solution, acetone, and isopropyl alcohol for 15 minutes, followed by drying with N_2 and treated with UV-ozone for 15 minutes. Subsequently, 3 nm of Cr and 50 nm of Au were thermally evaporated with a shadow mask as the source/drain electrodes. Evaporation rates are 0.1–0.2 Å s^{-1} and 0.5–1.5 Å s^{-1} for Cr and Au, respectively. The length and width of the channel are 20 μm and 200 μm, respectively. The substrates with the electrodes underwent an additional 15 minutes UV-ozone cleaning. Then, the solution of p(g2T-T), p(g2T-T) : DtFDA mixture, or Homo-gDPP : DtFDA mixture was spin-coated on the substrates at 3000 rpm for 20 s. The films based on p(g2T-T) : DtFDA mixture or Homo-gDPP : DtFDA mixture were then cross-linked by exposure to 365 nm UV light (450 mW cm^{-2}) for 2 minutes with a shadow mask, followed by a 3 s development in chloroform and blow drying under N_2 gun. Last, the solution of the Cin-Cell : DtFDA mixture was spin-coated on the substrates at 5000 rpm for 20 s. The layer was then cross-linked by exposure to 365 nm UV light (450 mW cm^{-2}) for 2 minutes with a different shadow mask, followed by a 3 s development in chloroform and drying with N_2. Note, that when the semiconductor layer is pure p(g2T-T), no patterning and encapsulation processes were conducted.

vOECT fabrication

vOECTs were also fabricated on pre-cleaned Si/100 nm SiO_2 wafers. First, 3 nm of Cr and 150 nm of Au were thermally evaporated with a shadow mask as the bottom source electrode. The width of the bottom source electrode is 30 μm. The substrates with the bottom electrodes underwent a 15 minutes UV-ozone cleaning process. Then, the solution containing p(g2T-T), p(g2T-T) : DtFDA mixture, or Homo-gDPP : DtFDA mixture was spin-coated on the substrates at 3000 rpm for 20 s. The films based on the p(g2T-T) : DtFDA mixture or Homo-gDPP : DtFDA mixture were then cross-linked by exposure to 365 nm UV light (450 mW cm^{-2}) for 2 minutes with a shadow mask, followed by a 3 s development in chloroform and drying with N_2. 150 nm of Au was then thermally evaporated with a shadow mask as the top drain electrode while maintaining the substrate at a temperature of approximately 5 °C with a back water-cooling system. The width of the top drain electrode is maintained at 30 μm. Subsequently, the solution of the Cin-Cell : DtFDA mixture was spin-coated on the substrates at 5000 rpm for 20 s. This layer was then cross-linked by exposure to 365 nm UV light (450 mW cm^{-2}) for 2 minutes with a different shadow mask, followed by a 3 s development in chloroform and drying with N_2. Note, that when the semiconductor layer is pure p(g2T-T), no patterning and encapsulation processes were conducted.

OECTs measurement

Electrical characterization of the OECTs was conducted with an FS-Pro (PDA) semiconductor parameter analyzer along with a probe station (Chengdu Chiptest Technology Co., LTD). To maintain a consistent PBS electrolyte concentration (0.01 M) during cycling stability measurements, a PDMS well was placed on the top of the device's active area and approximately 40 μL electrolyte was added. An Ag/AgCl electrode (World Precision Instruments, EP-1) was immersed in the electrolyte to act as the gate electrode.

Film characterization

AFM characterizations were acquired with a Bruker Dimension Icon System in tapping mode under ambient conditions. CV measurements were carried out with a Gamry Reference 620. 0.01 M PBS solution was utilized as the electrolyte, while a saturated Ag/AgCl electrode acted as the reference electrode, and a 10 mm × 10 mm platinum plate was employed as the counter electrode. Solution of p(g2T-T) or p(g2T-T) : DtFDA mixture was spin-coated on Cr (3 nm)/Au (50 nm) deposited glass substrates (5 mm × 5 mm) to make the working electrode. A scan rate of 50 mV s^{-1} was applied for the CV measurements.

Author contributions

M. X. was responsible for conceptualization, formal analysis, investigation, visualization, writing-original draft, and writing-review & editing. Y. L., M. L. was responsible for resources. D. Z., C. H., and L. B. were responsible for formal analysis and methodology. Y. C. was responsible for resources and project administration. J. C., L.-W. F. was responsible for resources and supervision. W. H. was responsible for conceptualization, resources, funding acquisition, methodology, project administration, and writing-review & editing.

Conflicts of interest

There are no conflicts to declare.

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References

1 A. Nawaz, Q. Liu, W. L. Leong, K. E. Fairfull-Smith and P. Sonar, *Adv. Mater.*, 2021, **33**, e2101874.

2 Z. Zhao, Z. Tian and F. Yan, *Cell Rep. Phys. Sci.*, 2023, **4**, 101673.

3 R. B. Rashid, X. Ji and J. Rivnay, *Biosens. Bioelectron.*, 2021, **190**, 113461.

4 L. Gao, M. Wu, X. Yu and J. Yu, *Int. J. Extreme Manuf.*, 2023, **6**, 012005.

5 T. Li, J. Y. Cheryl Koh, A. Moudgil, H. Cao, X. Wu, S. Chen, K. Hou, A. Surendran, M. Stephen, C. Tang, C. Wang, Q. J. Wang, C. Y. Tay and W. L. Leong, *ACS Nano*, 2022, **16**, 12049–12060.

6 R. B. Rashid, W. Du, S. Griggs, I. P. Maria, I. McCulloch and J. Rivnay, *Sci. Adv.*, 2021, **7**, eabh1055.

7 X. Wu, S. Chen, M. Moser, A. Moudgil, S. Griggs, A. Marks, T. Li, I. McCulloch and W. L. Leong, *Adv. Funct. Mater.*, 2022, **33**, 2209354.

8 Y. Jimbo, D. Sasaki, T. Ohya, S. Lee, W. Lee, F. Arab Hassani, T. Yokota, K. Matsuura, S. Umezu, T. Shimizu and T. Someya, *Proc. Natl. Acad. Sci. U. S. A.*, 2021, **118**, e2022300118.

9 K. Lieberth, P. Romele, F. Torricelli, D. A. Koutsouras, M. Bruckner, V. Mailander, P. Gkoupidenis and P. W. M. Blom, *Adv. Healthcare Mater.*, 2021, **10**, e2100845.

10 Y. Liang, F. Brings, V. Maybeck, S. Ingebrandt, B. Wolfrum, A. Pich, A. Offenhäusser and D. Mayer, *Adv. Funct. Mater.*, 2019, **29**, 1902085.

11 J. Ji, H. Wang, R. Liu, X. Jiang, Q. Zhang, Y. Peng, S. Sang, Q. Sun and Z. L. Wang, *Nano Energy*, 2021, **87**, 106116.

12 S. K. Lee, Y. W. Cho, J. S. Lee, Y. R. Jung, S. H. Oh, J. Y. Sun, S. Kim and Y. C. Joo, *Adv. Sci.*, 2021, **8**, 2001544.

13 Z. Xie, C. Zhuge, Y. Zhao, W. Xiao, Y. Fu, D. Yang, S. Zhang, Y. Li, Q. Wang, Y. Wang, W. Yue, I. McCulloch and D. He, *Adv. Funct. Mater.*, 2022, **32**, 2107314.

14 Y. Wang, S. Wustoni, J. Surgailis, Y. Zhong, A. Koklu and S. Inal, *Nat. Rev. Mater.*, 2024, **9**, 249–265.

15 J. Rivnay, S. Inal, A. Salleo, R. M. Owens, M. Berggren and G. G. Malliaras, *Nat. Rev. Mater.*, 2018, **3**, 17086.

16 W. Huang, J. Chen, G. Wang, Y. Yao, X. Zhuang, R. M. Pankow, Y. Cheng, T. J. Marks and A. Facchetti, *J. Mater. Chem. C*, 2021, **9**, 9348–9376.

17 D. Khodagholy, J. Rivnay, M. Sessolo, M. Gurfinkel, P. Leleux, L. H. Jimison, E. Stavrinidou, T. Herve, S. Sanaur, R. M. Owens and G. G. Malliaras, *Nat. Commun.*, 2013, **4**, 2133.

18 X. Wu, Q. Liu, A. Surendran, S. E. Bottle, P. Sonar and W. L. Leong, *Adv. Electron. Mater.*, 2020, **7**, 2000701.

19 L. Q. Flagg, C. G. Bischak, J. W. Onorato, R. B. Rashid, C. K. Luscombe and D. S. Ginger, *J. Am. Chem. Soc.*, 2019, **141**, 4345–4354.

20 L. Q. Flagg, R. Giridharagopal, J. Guo and D. S. Ginger, *Chem. Mater.*, 2018, **30**, 5380–5389.

21 A. A. Szumska, I. P. Maria, L. Q. Flagg, A. Savva, J. Surgailis, B. D. Paulsen, D. Moia, X. Chen, S. Griggs, J. T. Mefford, R. B. Rashid, A. Marks, S. Inal, D. S. Ginger, A. Giovannitti and J. Nelson, *J. Am. Chem. Soc.*, 2021, **143**, 14795–14805.

22 H. W. Kim, V. J. Bukas, H. Park, S. Park, K. M. Diederichsen, J. Lim, Y. H. Cho, J. Kim, W. Kim, T. H. Han, J. Voss, A. C. Luntz and B. D. McCloskey, *ACS Catal.*, 2019, **10**, 852–863.

23 E. Mitraka, M. Gryszel, M. Vagin, M. J. Jafari, A. Singh, M. Warczak, M. Mitrakas, M. Berggren, T. Ederth, I. Zozoulenko, X. Crispin and E. D. Głowacki, *Adv. Sustainable Syst.*, 2018, **3**, 1800110.

24 A. Giovannitti, R. B. Rashid, Q. Thiburce, B. D. Paulsen, C. Cendra, K. Thorley, D. Moia, J. T. Mefford, D. Hanifi, D. Weiyuan, M. Moser, A. Salleo, J. Nelson, I. McCulloch and J. Rivnay, *Adv. Mater.*, 2020, **32**, e1908047.

25 M. Moser, T. C. Hidalgo, J. Surgailis, J. Gladisch, S. Ghosh, R. Sheelamanthula, Q. Thiburce, A. Giovannitti, A. Salleo, N. Gasparini, A. Wadsworth, I. Zozoulenko, M. Berggren, E. Stavrinidou, S. Inal and I. McCulloch, *Adv. Mater.*, 2020, **32**, e2002748.

26 I. P. Maria, B. D. Paulsen, A. Savva, D. Ohayon, R. Wu, R. Hallani, A. Basu, W. Du, T. D. Anthopoulos, S. Inal, J. Rivnay, I. McCulloch and A. Giovannitti, *Adv. Funct. Mater.*, 2021, **31**, 2008718.

27 M. Moser, L. R. Savagian, A. Savva, M. Matta, J. F. Ponder, T. C. Hidalgo, D. Ohayon, R. Hallani, M. Reisjalali, A. Troisi, A. Wadsworth, J. R. Reynolds, S. Inal and I. McCulloch, *Chem. Mater.*, 2020, **32**, 6618–6628.

28 D. Ohayon, L. Q. Flagg, A. Giugni, S. Wustoni, R. Li, T. C. Hidalgo Castillo, A.-H. Emwas, R. Sheelamanthula, I. McCulloch, L. J. Richter and S. Inal, *ACS Mater. Au*, 2023, **3**, 242–254.

29 T. C. Hidalgo Castillo, M. Moser, C. Cendra, P. D. Nayak, A. Salleo, I. McCulloch and S. Inal, *Chem. Mater.*, 2022, **34**, 6723–6733.

30 M. Stephen, X. Wu, T. Li, T. Salim, K. Hou, S. Chen and W. L. Leong, *Mater. Horiz.*, 2022, **9**, 2408–2415.

31 A. Håkansson, S. Han, S. Wang, J. Lu, S. Braun, M. Fahlman, M. Berggren, X. Crispin and S. Fabiano, *J. Polym. Sci., Part B: Polym. Phys.*, 2017, **55**, 814–820.

32 T. Nguyen-Dang, K. Harrison, A. Lill, A. Dixon, E. Lewis, J. Vollbrecht, T. Hachisu, S. Biswas, Y. Visell and T.-Q. Nguyen, *Adv. Electron. Mater.*, 2021, **7**, 2100519.

33 A. Weissbach, L. M. Bongartz, M. Cucchi, H. Tseng, K. Leo and H. Kleemann, *J. Mater. Chem. C*, 2022, **10**, 2656–2662.

34 M. Azimi, C. H. Kim, J. Fan and F. Cicoira, *Faraday Discuss.*, 2023, **246**, 540–555.

35 M. Azimi, A. Subramanian, J. Fan, F. Soavi and F. Cicoira, *J. Mater. Chem. C*, 2023, **11**, 4623–4633.

36 C. H. Kim, M. Azimi, J. Fan, H. Nagarajan, M. Wang and F. Cicoira, *Nanoscale*, 2023, **15**, 3263–3272.

37 M. Xie, H. Liu, M. Wu, C. Chen, J. Wen, L. Bai, J. Yu and W. Huang, *Org. Electron.*, 2023, **117**, 106777.

38 X. Wang, Z. Zhang, P. Li, J. Xu, Y. Zheng, W. Sun, M. Xie, J. Wang, X. Pan, X. Lei, J. Wang, J. Chen, Y. Chen, S. J. Wang and T. Lei, *Adv. Mater.*, 2024, 2400287.

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39 E. Zeglio, J. Eriksson, R. Gabrielsson, N. Solin and O. Inganas, *Adv. Mater.*, 2017, **29**, e1807393.
40 S. T. Keene, L. W. Gatecliff, S. L. Bidinger, M. Moser, I. McCulloch and G. G. Malliaras, *MRS Commun.*, 2023, **14**, 158–166.
41 T. J. Quill, G. LeCroy, A. Marks, S. A. Hesse, Q. Thiburce, I. McCulloch, C. J. Tassone, C. J. Takacs, A. Giovannitti and A. Salleo, *Adv. Mater.*, 2024, e2310157.
42 Y.-Q. Zheng, Y. Liu, D. Zhong, S. Nikzad, S. Liu, Z. Yu, D. Liu, H.-C. Wu, C. Zhu, J. Li, H. Tran, J. B.-H. Tok and Z. Bao, *Science*, 2021, **373**, 88–94.
43 A. Savva, C. Cendra, A. Giugni, B. Torre, J. Surgailis, D. Ohayon, A. Giovannitti, I. McCulloch, E. Di Fabrizio, A. Salleo, J. Rivnay and S. Inal, *Chem. Mater.*, 2019, **31**, 927–937.
44 S. Zhang, P. Ding, T. P. Ruoko, R. Wu, M. A. Stoeckel, M. Massetti, T. Liu, M. Vagin, D. Meli, R. Kroon, J. Rivnay and S. Fabiano, *Adv. Funct. Mater.*, 2023, **33**, 2302249.
45 A. F. Paterson, H. Faber, A. Savva, G. Nikiforidis, M. Gedda, T. C. Hidalgo, X. Chen, I. McCulloch, T. D. Anthopoulos and S. Inal, *Adv. Mater.*, 2019, **31**, e1902291.
46 W. Huang, J. Chen, Y. Yao, D. Zheng, X. Ji, L. W. Feng, D. Moore, N. R. Glavin, M. Xie, Y. Chen, R. M. Pankow, A. Surendran, Z. Wang, Y. Xia, L. Bai, J. Rivnay, J. Ping, X. Guo, Y. Cheng, T. J. Marks and A. Facchetti, *Nature*, 2023, **613**, 496–502.
47 J. Kim, R. M. Pankow, Y. Cho, I. D. Duplessis, F. Qin, D. Meli, R. Daso, D. Zheng, W. Huang, J. Rivnay, T. J. Marks and A. Facchetti, *Nat. Electron.*, 2024, **7**, 234–243.
48 M. J. Donahue, A. Williamson, X. Strakosas, J. T. Friedlein, R. R. McLeod, H. Gleskova and G. G. Malliaras, *Adv. Mater.*, 2018, **30**, 1705031.
49 S. Y. Jeong, J. W. Moon, S. Lee, Z. Wu, S. H. Park, J. H. Cho and H. Y. Woo, *Adv. Electron. Mater.*, 2023, **9**, 2300053.
50 Z. Wang, X. Zhuang, Y. Chen, B. Wang, J. Yu, W. Huang, T. J. Marks and A. Facchetti, *Chem. Mater.*, 2019, **31**, 7608–7617.
51 C. B. Nielsen, A. Giovannitti, D. T. Sbircea, E. Bandiello, M. R. Niazi, D. A. Hanifi, M. Sessolo, A. Amassian, G. G. Malliaras, J. Rivnay and I. McCulloch, *J. Am. Chem. Soc.*, 2016, **138**, 10252–10259.